



Project Overview

CipherWing is an offline-first file encryption and decryption application that runs directly inside browser memory. By leveraging web standards, it provides high-strength data protection without uploading data to external servers, delivering 100% client-side privacy.

Problem Statement

Existing web-based file encryptors pose substantial data leakage risks by uploading source files to backend servers for processing. Additionally, many tools use outdated, insecure ciphers (e.g. DES) and suffer from strict file-size constraints (10-15MB) due to naive in-memory loading limits.

Objective & Stack

To construct a standalone, high-performance offline suite capable of processing files up to 2GB+ using chunked streaming, while isolating user sessions via IndexedDB vaults.

Cryptography
Web Crypto
API

Local Database
IndexedDB

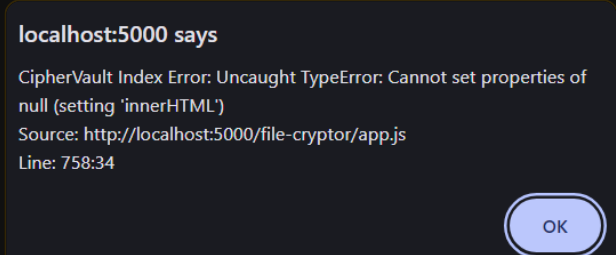
Graphics 3D
Three.js
WebGL

Key Derivation
PBKDF2 (600k)

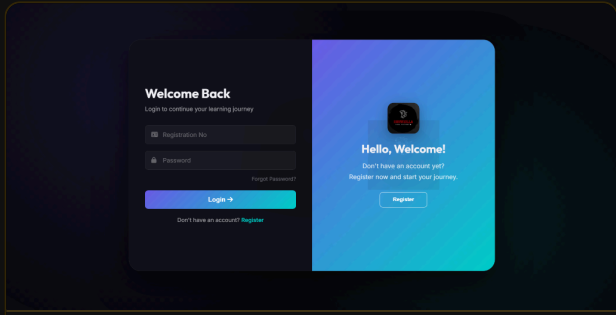
System Architecture



Working Screenshots



Workspace Dashboard with Yellow/Black Theme



Interactive File Queue & Progress Indicators

Core Modules

Local Auth

User account hashing via PBKDF2 (600k iterations) inside IndexedDB browser store.

Stream Worker

Slices large files into 1MB chunks to process 2GB+ files off-thread without tab crashes.

Three.js UI

Interactive 3D wireframe padlock background reacting to mouse move parallax.

DB History

Scoped profiles history log with storage metrics and instant local re-downloads.

System Metrics

- ✓ **0% Data Leaks:** Zero server communication ensures full privacy.
- ✓ **2GB+ Chunking Limit:** Seamless 1MB slice streaming.
- ✓ **PBKDF2 Security:** 600,000 hashing iterations matching OWASP standards.
- ✓ **Dual Theme Sync:** Seamless WebGL color switching between dark & light themes.

Future Scope

Future developments focus on implementing WebRTC P2P direct transfers to share encrypted files directly between browser tabs without intermediate storage servers, and integrating multi-threaded Web Workers for parallel batch processing.

